



AUDITING REPRESENTATION AND EQUITY IN SMART CITY AI DATASETS

A Data-Driven Audit of Demographic Representation, Transportation Behaviour and Urban Accessibility



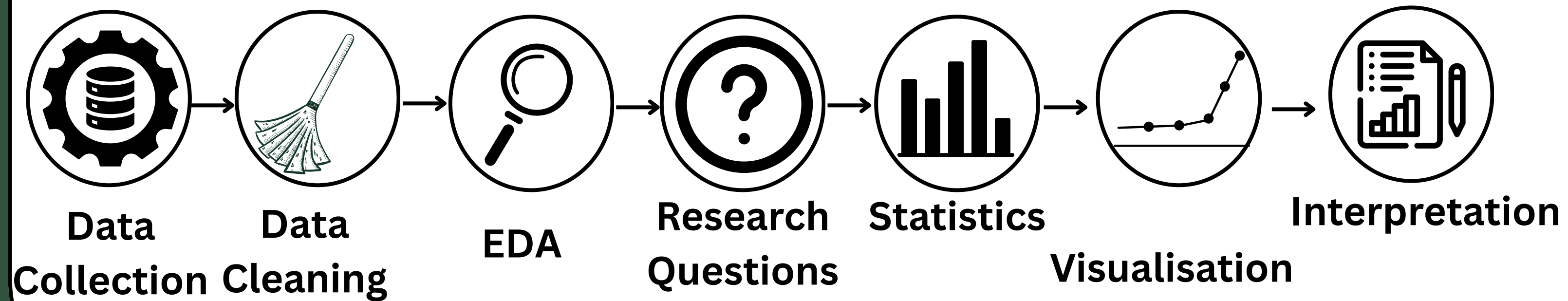
Scan for
Code
and Data

01 INTRODUCTION

Smart City AI systems increasingly rely on citizen-generated data for urban planning, mobility management and public services.

This study audits Smart City datasets to evaluate representation, behaviour and accessibility equity.

03 METHODOLOGY WORKFLOW



02 DATASETS

Dataset 1:
smart_city_citizen_activity.csv

- Age
- Gender
- Transportation Mode
- Behaviour Activities

Dataset 2:
urban_connectivity.csv

- Accessibility
- Income Groups
- Demographic Groups
- Urban Metrics

04 RESEARCH QUESTIONS

RQ 1
Who is represented in Smart City AI datasets?

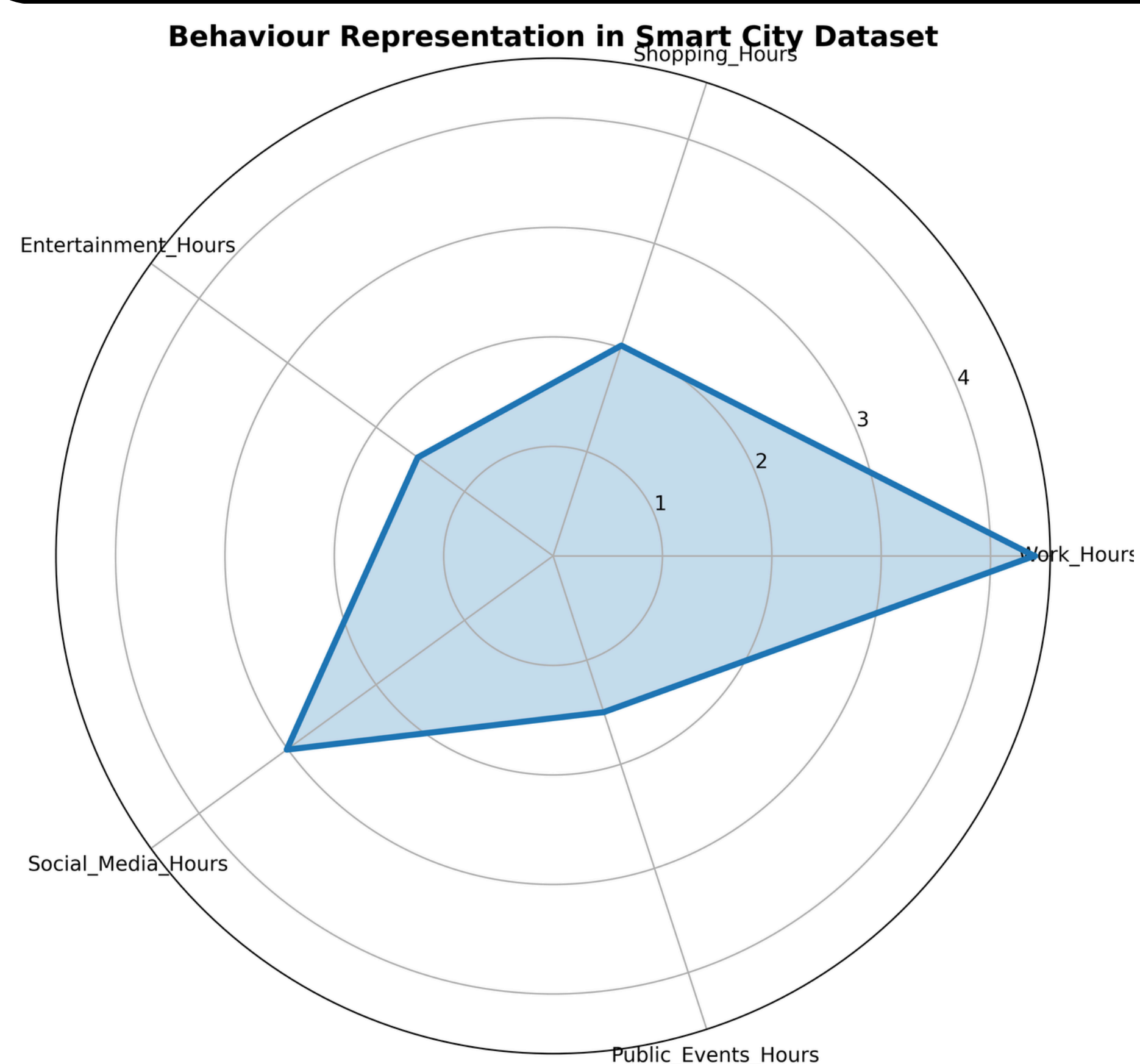
RQ 2
Whose Transportation behaviours are captured?

RQ 3
Are demographic groups equally served by urban infrastructure?

RQ 4
Does income influence urban accessibility and representation?

RQ 5
Which citizen behaviours dominate the Smart City dataset?

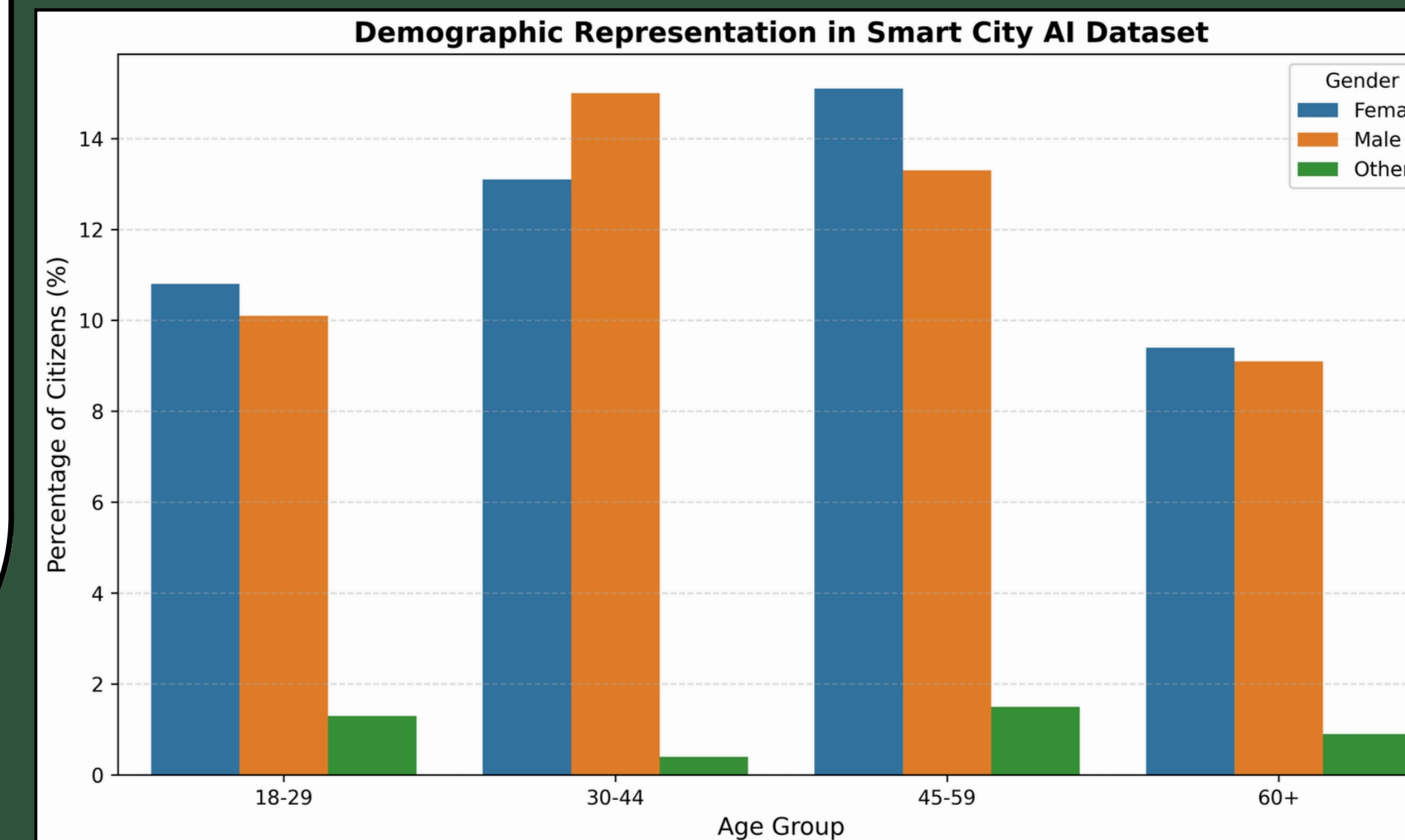
RQ5: Behaviour Representation in Smart City Dataset



Work-related activity has the highest average time, followed by social media usage. Public events and entertainment are least represented.

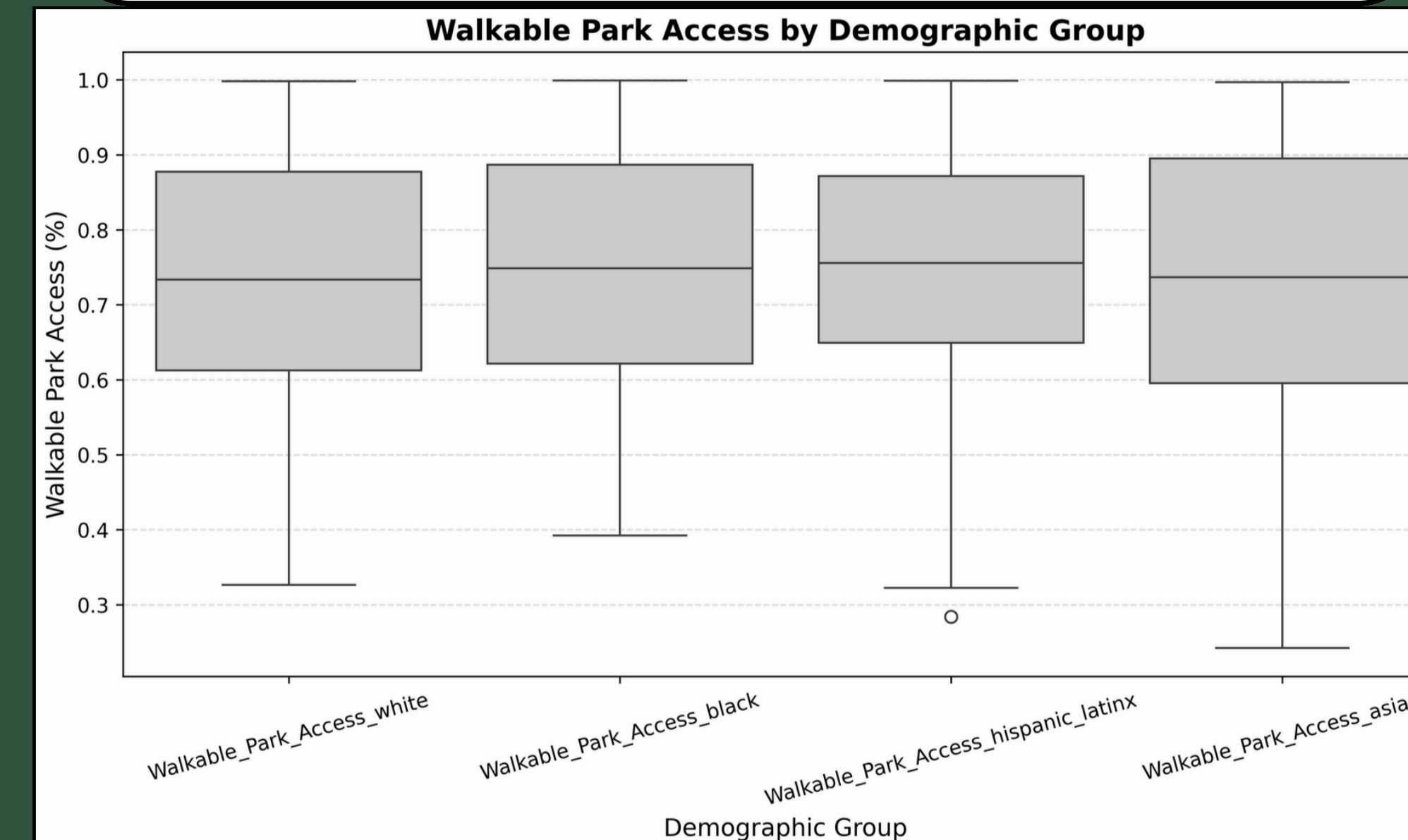
05 RESULTS AND FINDINGS

RQ1: Demographic Representation Across Age Groups



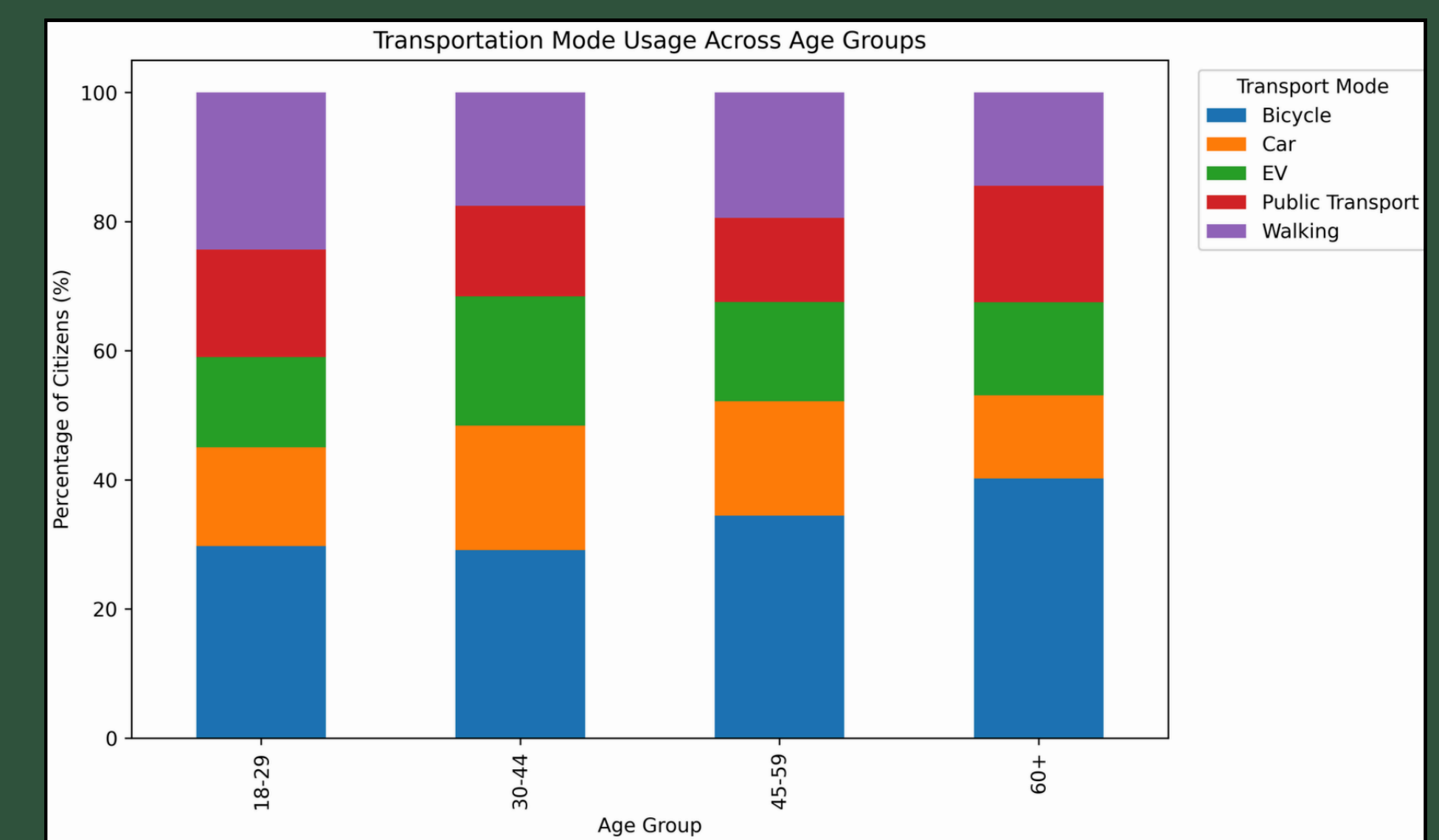
Gender representation is relatively balanced across age groups, with the 'Other' category showing minimal representation.

RQ 3: Are demographic groups equally served by urban infrastructure?



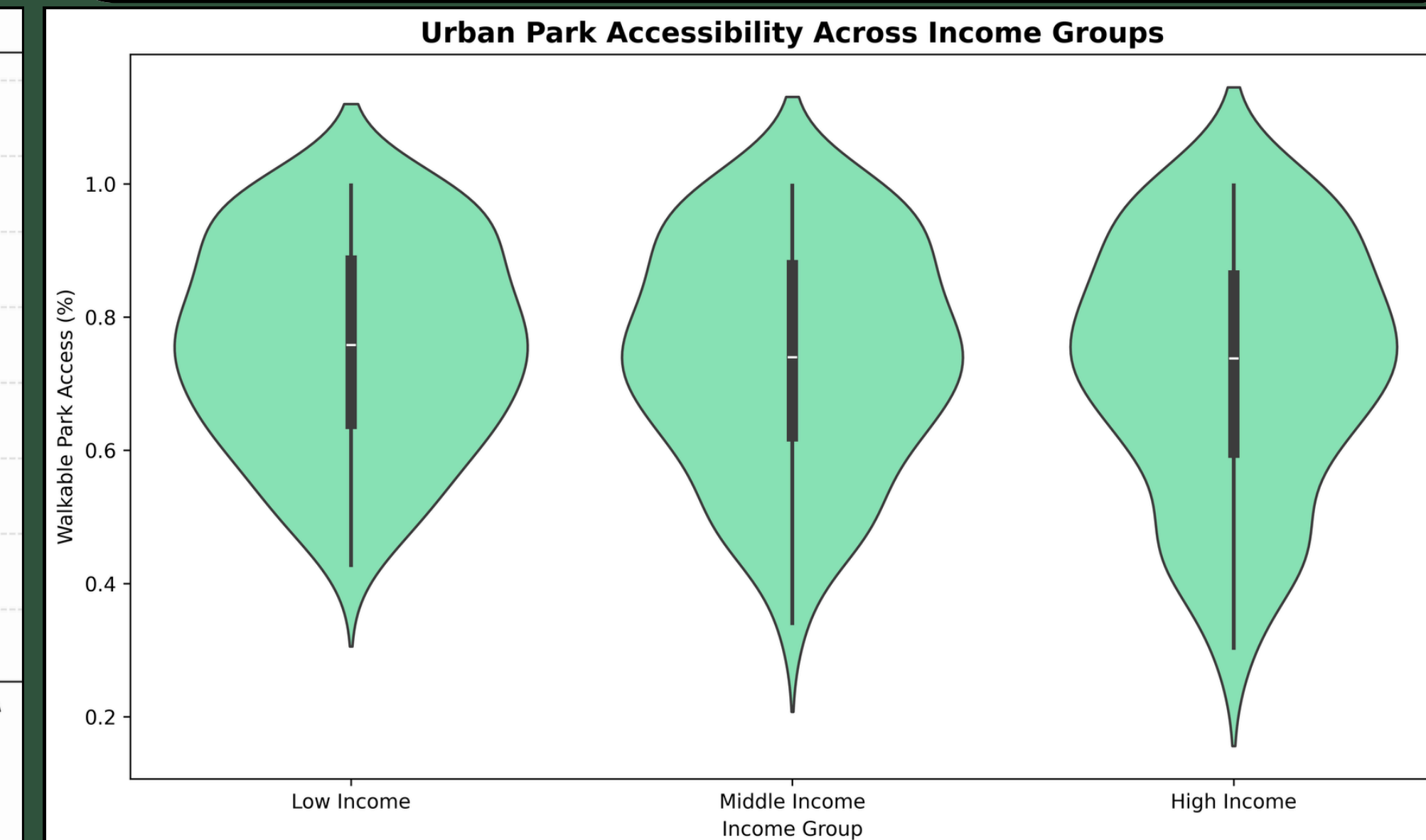
Walkable park access distributions are similar across demographic groups, with median accessibility around 70-75%.

RQ 2: Transportation Mode Usage Across Age Groups



Walking and cycling are the most preferred transportation modes across all age groups, especially among younger citizens.

RQ 4: Urban Park Accessibility Across Income Groups



Accessibility levels are comparable across income groups indicating limited income-based disparity.

06 CONTRIBUTIONS

- ✓ Comprehensive audit of demographic representation
- ✓ Transportation behaviour analysis across age groups
- ✓ Accessibility equity assessment
- ✓ Behavioural pattern analysis
- ✓ Reproducible Smart City AI auditing framework

07 CONCLUSION

This study provides an empirical audit of Smart City AI datasets across demographic representation, transportation behaviour, park accessibility and behavioural patterns.

Results indicate balanced gender representation but a strong dominance of work-related activities. Accessibility remains relatively consistent across demographic and income groups.

These findings can guide city authorities and AI developers toward fairer, more inclusive and representative Smart City systems.

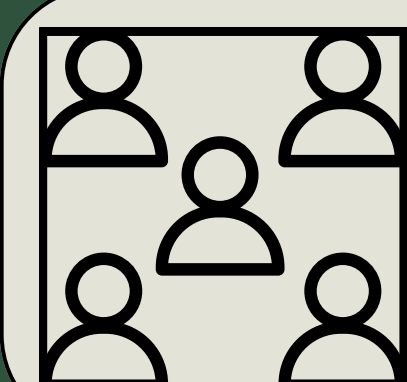
08 IMPLICATIONS

- AI models trained on imbalanced behavioural data may over-prioritise certain activities.
- Cities should encourage diverse data collection practices.
- Continuous auditing is essential to ensure equitable and transparent Smart City AI systems.

References

- OECD (2020). Smart Cities and Inclusive Growth.
- UN-Habitat (2022). World Cities Report.
- IEEE Smart Cities Initiative (2023).

KEY TAKEAWAYS



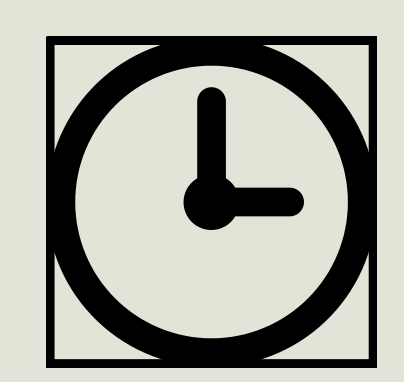
Balanced demographic representation across age and gender.



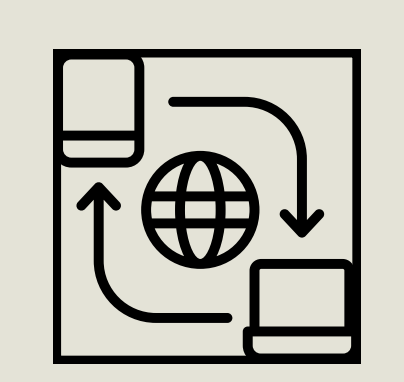
Walking and Cycling are the most preferred transportation mode.



Income does not significantly influence park accessibility.



Work related activities dominate behavioural representation.



Park Accessibility is similar across demographic groups.